

From 50MHz to 100GHz, Bench-Top to Wristwatch Oscilloscopes Have Come a Long Way

In this story, we explore the latest features and kinds of oscilloscopes that play an integral role in effectively testing designs today. We also learn how new technologies in oscilloscopes are enabling test and embedded engineers to design better

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As an engineer, if you want to improve your design overall, effective testing is an integral part. However, it means possessing tools you can rely on and that can provide consistent results as per requirements.

Today, oscilloscopes come in numerous shapes and sizes—from as small as a wristwatch to pocket-sized and bench-top ones. They even have diverse bandwidths—from the entry-level 50MHz to the massive 100GHz! Let us explore the new technologies, innovations and breakthroughs in the oscilloscopes domain that help our engineers design better.

From analogue to digital

From simple signal capture to complex time-domain signal analysis, oscilloscopes come with salient and contrasting functionalities. Oscilloscope manufacturers have augmented the automated capabilities and portability factor so that designers can accomplish swiftly and get more authentic measurements for increased productivity.

The invention of cathode ray tubes (CRTs) in the late 19th century was an important milestone for the birth of analogue oscilloscopes, and has changed the way of waveform representation, believes Srinivasa Appalla, manager-product support & application, Rohde & Schwarz. However, he says, “The technological advancements in electronic designs have put new requirements on signal analysis which are not met by analogue scopes. Digital oscilloscopes, introduced in the 80s, are dominating the market now because of the ability of

Exciting innovations in the last one year



Oscium's IMSO-204 and IMSO-204L are mixed-signal oscilloscopes designed specifically for the iPhone, iPad and iPod.

Highlights: 2 analogue + 4 digital channels, 50MSPS sample rate, 5MHz bandwidth, 200ns/div-10s/div

Price: \$399.97

Gabotronics' Xminilab Portable is a small mixed-signal oscilloscope with an arbitrary waveform generator and protocol sniffer.

Highlights: 2 analogue inputs, maximum sampling rate of 2MSPS, 200kHz analogue bandwidth, 8-bit resolution, 1MΩ 15pF input impedance, 256 buffer size per channel and input voltage range of -14V to +20V

Price: \$118



LabNation's SmartScope is a 100MS/s open source oscilloscope for iPad, Android and PC. A must-have for every Arduino and Raspberry Pi developer.

Highlights: 2x100MS/s 45MHz oscilloscope, 50MS/s arbitrary waveform generator, digital logic analyser at 100MS/s, digital waveform generator at 100MS/s, 200 waveforms/second data updates

Price: \$179

Gabotronics' Oscilloscope Watch

Highlights: 2 analogue inputs, 4MSPS maximum sampling rate, 200kHz analogue bandwidth, 8-bit resolution

Price: \$150



digital signal processing, and respective analysis and documentation.”

Sumit Sharma, marketing manager-India, Good Will Instrument Co. Ltd. too believes, that with the rapid advancement of technology, the oscilloscope market has also been shifting from conventional analogue oscilloscopes towards digital storage oscilloscopes (DSOs). He says, “In contrast to analogue oscilloscopes, the major function of a DSO is to not only convert signals from analogue to digital but also store testing data, allow remote control and transmission of data through various interfaces.” But, he also believes, “Despite the strengths of DSOs, analogue oscilloscopes still play

an important role of providing real-time signals and waveform display.”

Mixed-signal and mixed-domain

“As oscilloscopes have evolved from analogue to digital and from mixed-signal oscilloscopes to mixed-domain oscilloscopes, they have become even more valuable tools for analysing system performance or troubleshooting problems,” says Pamela Aparo, marketing manager-electronic test & measurement, Analog Devices Inc.

Embedded system designers require advanced testing capabilities to resolve their design issues, so mixed-signal oscilloscopes (MSOs) emerged in the